

Third Semester B. Tech. (Electronics and Communication Engineering) Examination

NETWORK THEORY

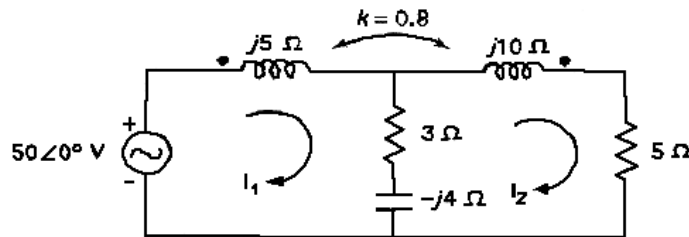
Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

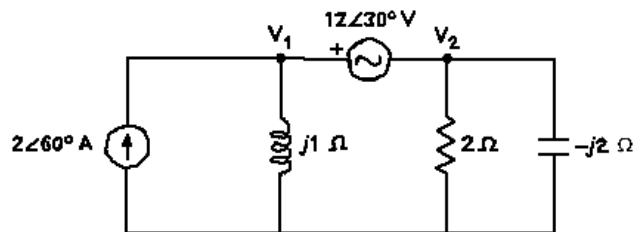
- (1) All questions are compulsory.
- (2) All questions carry marks as indicated against them.
- (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. (a) Determine Voltage across $5\ \Omega$ resistor in following network.



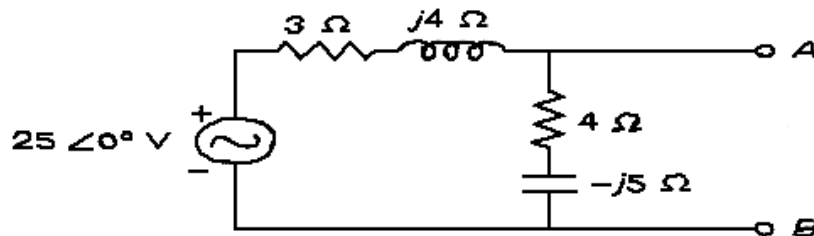
5(CO1)

- (b) Compute V_2 in following network.



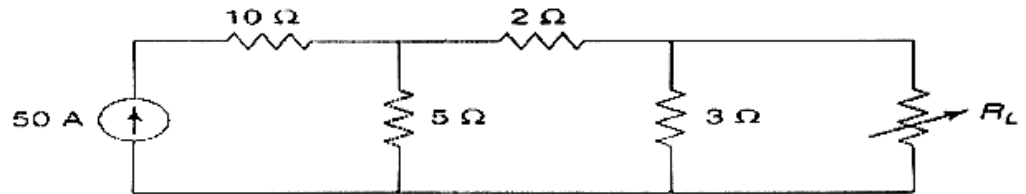
5(CO1)

2. (a) Obtain Norton's equivalent network between terminal A and B.



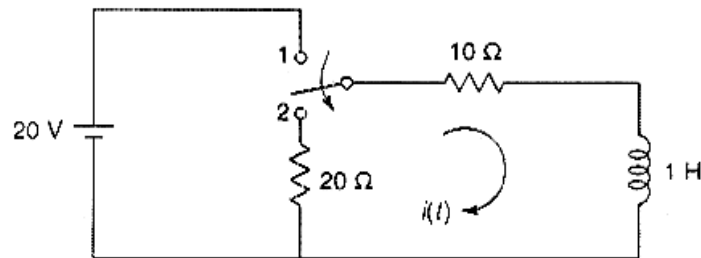
5(CO3)

- (b) Determine the value of R_L for maximum power transfer along with maximum power.



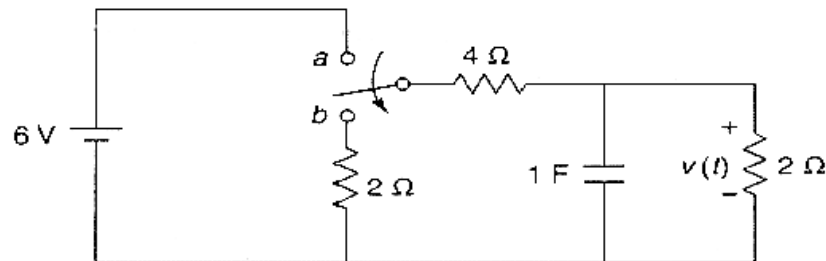
5(CO3)

3. (a) In the network shown in Figure, the switch is changed from the position 1 to the position 2 at $t=0$, steady condition having reached before switching. Compute i , di/dt and d^2i/dt^2 at $t=0+$.



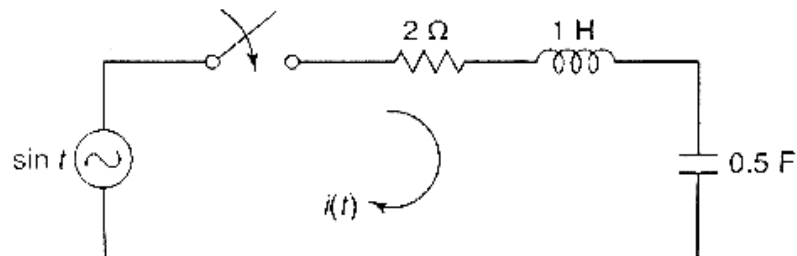
5(CO2)

- (b) In the network of Figure, the switch is moved from a to b, at $t=0$. Estimate $v(t)$.



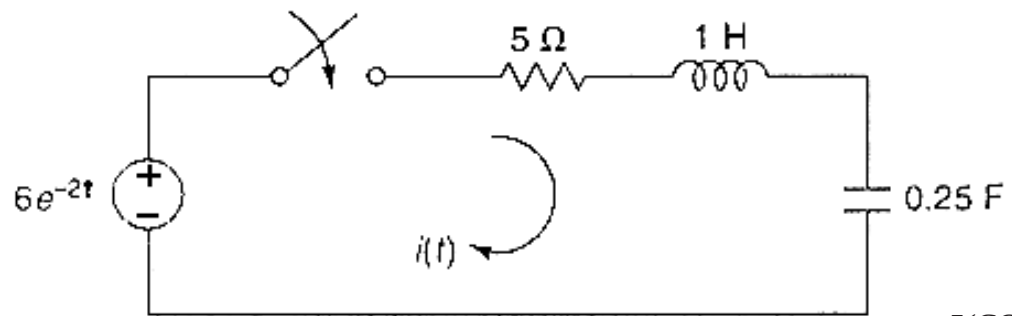
5(CO2)

4. (a) For the network shown in Figure, the switch is closed at $t=0$. Determine the current $i(t)$ assuming zero initial conditions.



5(CO2)

- (b) For the network shown in Figure, the switch is closed at $t=0$. Compute the current $i(t)$ assuming zero initial conditions in the network elements.



5(CO2)

5. (a) Obtain the pole-zero plot and classify the following functions :

(a) $F(s) = \frac{s(s+2)}{(s+1)(s+3)}$

(b) $F(s) = \frac{s(s+1)}{(s+2)^2(s+3)}$

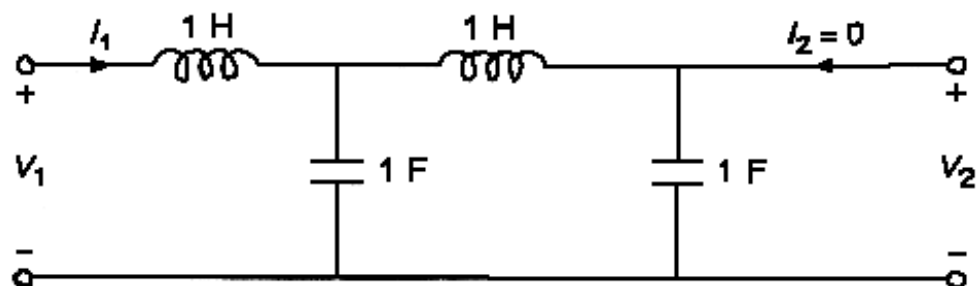
(c) $F(s) = \frac{s(s+2)}{(s+1+j1)(s+1-j1)}$

(d) $F(s) = \frac{(s+1)^2(s+5)}{(s+2)(s+3+j2)(s+3-j2)}$

(e) $F(s) = \frac{s^2+4}{(s+2)(s^2+9)}$

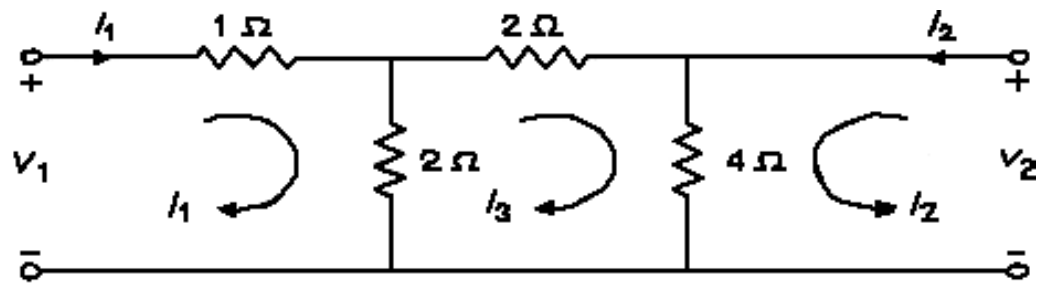
5(CO4)

- (b) Illustrate the network functions V_1/I_1 , V_2/V_1 and V_2/I_1 , for the network shown in Figure.



5(CO5)

6. (a) Estimate hybrid parameters for the network of Figure.



5(CO5)

- (b) Design a constant-k, T section low-pass filter is having 2.5kHz cut-off frequency and nominal impedance of 700Ω . Also compute the frequency at which this filter produces an attenuation of 19.1 dB. Illustrate its characteristic impedances and phase constant at pass band and stop or attenuation band. 5(CO3)

